Abbott LVAD Imaging Microscopy

Advaita Tikoti, Anaya Bawiskar, Samika Gandhi

Left Ventricular Assist Device (also referred to as LVAD) is a device used to assist weak left ventricles in pumping blood to the body. It is often implanted in patients with heart failure because their left ventricle cannot generate enough force to pump blood to the body, leading to low oxygen levels in organs, shortness of breath, regurgitation and low blood pressure [1].

Heartmate 3 by Abbott Cardiovascular employs an innovative magnetic-levitation technology with three crucial components - inflow cannula, magnetic rotor and outflow cannula. As shown in **Figure 1**, the inflow cannula diverts blood flow from the left ventricle and pushes it forward with the help of a magnetic rotor which exits through the outflow graft, opening into the aorta for the blood to be pumped to the body [2].

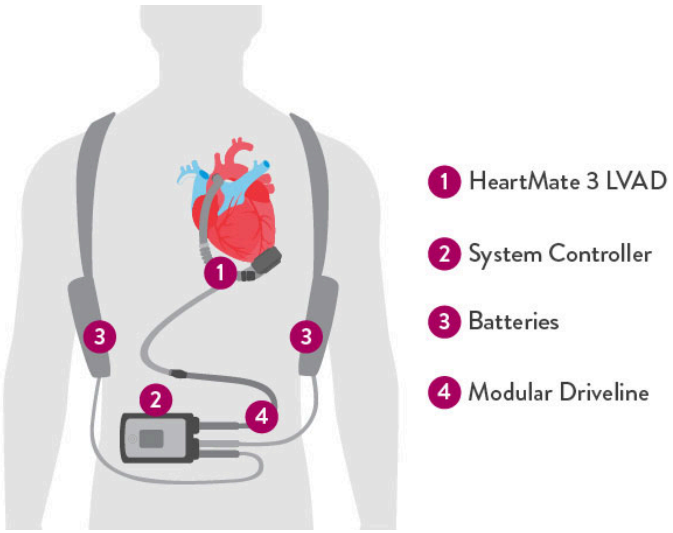


Figure 1 - HeartMate 3 placement and components [2]

Imaging Goals

The goal of this project is to assess the hemocompatibility, specifically hemolytic and thrombogenic responses, and overall safety of Abbott Cardiovascular's HeartMate 3, across its preclinical development pipeline. LVAD implantation is associated with several complications arising from blood-device interactions, including bleeding, infection, pump thrombosis, and device malfunction. These complications have significant consequences, such as reoperation or blood transfusion, and can increase the mortality and worsen long-term outcomes for patients [3]. While developing a novel LVAD, minimizing potential device-related complications is of utmost importance. Thorough pre-clinical testing of device functionality and biological interactions, both *in vitro* and *in vivo*, can help establish safety and effectiveness prior to clinical translation.

The study will adhere to International Organization for Standardization (ISO) 10993-4, which outlines the selection of tests for device interactions with blood. **Table 1** summarizes relevant pre-clinical tests in ISO 10993-4 for ventricular-assist devices, including hemolysis, thrombosis, platelet activation, hematologic response, complement activation, and gross surface analysis.

Table 1. ISO 10993-4 testing for ventricular-assist devices [4]

Category	**Type**	**Test(s)**	***In vitro/ In vivo***
Hemolysis	Material-induced	ASTM F756	*In vitro*
	Mechanically-induced	NIH (Normalized Index of Hemolysis), plasma free hemoglobin	*In vitro, in vivo*
Thrombosis	Coagulation	Thrombin & fibrin	*In vitro, in vivo*
	Platelet Activation	Count, activation, morphology (SEM)	*In vitro, in vivo*
	Hematology	WBC, platelet counts	*In vitro, in vivo*
	Complement system	Complement protein levels	*In vitro, in vivo*
	Gross / surface	Light microscopy, SEM, % occlusion, gross analysis	*In vitro, in vivo*

The first phase of the study will evaluate the *in vitro* hemocompatibility of the device under controlled physiologic blood flow conditions using a rabbit blood flow-loop model at 37°C and 4.5 L/min for 6 hours, following the flow-loop test setup described by ASTM F1841 [5][6]. As indicated in the standard, paired testing will be conducted with a legally marketed comparison device with the sample blood pool in a matched loop so that relative hemolysis comparisons can be made. Brightfield microscopy is used for assessment of blood samples collected at baseline and every 30 minutes, with Wright-Giesma-stained blood smears evaluated for red blood cell integrity, platelet aggregation, and white blood cell activation. This approach enables quantification of mechanically induced hemolysis and early inflammatory activation under physiologic shear stress in accordance with ASTM F756 and ISO 10993-4 guidelines [7][4].

At the endpoint, scanning electron microscopy (SEM) will be utilized on the device surface to visualize platelet, white blood cell (WBC), and red blood cell (RBC) adhesion and fibrous deposition, key indicators of thrombosis for ISO 10993-4. SEM analysis enables quantitative assessment of surface thrombogenicity through measurement of platelet coverage, aggregation, morphology, and early thrombus formation at the biomaterial interface.

Micro-computed tomography (micro-CT) will be used to quantify three-dimensional thrombus deposition on LVAD components following flow-loop exposure. Contrast-enhanced micro-CT will enable volumetric assessment of thrombus, spatial distribution, and percent occlusion across complex device geometry, providing information not captured by surface-based imaging techniques alone [8].

Overall, the imaging goal of the first half of the study is to quantitatively assess hemolysis, platelet activation, and thrombus formation across multiple scales, ensuring that levels of hemolysis and thrombogenicity fall within acceptable ranges *in-vitro* before proceeding to *in-vivo* testing in an animal model. Acceptable hemolysis thresholds as defined per ASTM F756 are: non-hemolytic (< 2%) , slightly hemolytic (2-5%), and hemolytic (> 5%). This device must remain in the non-hemolytic range (<2%) [7].

In the second half of the study, the device will be implanted via thoracotomy in an ovine model (n = 10) for 90 days to assess *in vivo* hemocompatibility, thrombogenicity, and overall device safety [9]. The sample size and duration is consistent with preclinical animal studies outlined in Summary of Safety and Effectiveness (SSED) data for similar FDA-cleared LVADs [10], while the ovine model demonstrates comparable vascular physiology to humans [9]. Throughout the study duration, periodic blood samples will be collected at baseline, post-operative day 1, and at regular weekly intervals to monitor hematological parameters such as RBC count, WBC count, platelet count, hemoglobin levels, and markers of hemolysis.

To enable non-invasive, longitudinal assessment of device performance and thrombogenicity, cardiac computed tomography (CT) angiography with intravenous contrast will be performed at defined timepoints (baseline post-implantation, 1 month, and 3 months) to assess both acute and progressive device-related changes. CT imaging will be used to evaluate device positioning, inflow cannula and outflow graft geometry, as well as detect complications such as thrombus formation, graft kinking, or obstruction [11][12]. Quantification will include thrombus volume, occlusion, and cross-sectional area. Inflow cannula angulation and graft geometry will be measured to detect malpositioning or kinking [13]. CCT will enable high-resolution anatomical assessment to detect structural abnormalities such as thrombus formation and graft obstruction to support evaluation of device safety and performance.

Upon device explantation at the 90-day endpoint, brightfield microscopy and hematoxylin and eosin (H&E) staining will be used for histopathological evaluation of tissues surrounding the implantation as well as distal organs such as the kidneys and liver. Analysis will focus on identifying inflammation, fibrosis, thrombus formation, and embolism, consistent with FDA guidance for cardiovascular device evaluation. Then, SEM will be used to evaluate surface thrombus formation and cell adhesion on the device surface, analyzing percent surface coverage and thrombus density, corresponding to thrombogenicity endpoints in ISO 10993-4 [4].

Overall, the imaging goals of this part will be used to verify that in vivo performance remains within the following quantitative success metrics:

- Hemolysis: Plasma free hemoglobin < 20 mg/dL and NIH < 0.01 g/100 L (ASTM F756 compliant) [7]
- Platelet adhesion: $< 5\%$ surface coverage on SEM analysis
- Thrombus formation: $< 10\%$ device surface occlusion
- RBC morphology: $< 5\%$ abnormal cells observed per field
- WBC activation: No significant clustering compared to control ($p > 0.05$)

Successful preclinical validation will support progression towards human clinical studies.

Imaging Testing Protocols

In vitro study

A flow-loop with fresh rabbit blood at 37°C and physiologic shear conditions will be set up to test the HeartMate 3 LVAD, following the setup in ASTM F1841 [6]. Blood contacting components will first be cleaned according to the manufacturer protocol before being exposed to rabbit blood under controlled conditions.The flow loop will be run at a rate of 4.5L/min for 6 hours [5], allowing continuous exposure of blood to the device under controlled hemodynamic conditions. Paired testing will be conducted between the HeartMate III and an FDA-cleared LVAD to assess relative hemocompatibility, with primary focus on hemolysis and cell activation. The use of mock circulatory loops are supported by ISO 14708-5, which outlines safety and performance requirements for active implantable circulatory support devices [14].

Brightfield Microscopy

1. Justification

Brightfield microscopy is used to provide rapid assessment of RBC morphology, platelet aggregation, and WBC activation in blood samples collected during the flow-loop study. This method supports evaluation of mechanically induced hemolysis and early cellular responses under physiologic shear conditions, testing indicated for LVADs in ISO 10993-4 [4]. Its use is aligned with ASTM F1841, which defines in vitro flow-loop testing for blood-contacting devices, and ASTM F756, which establishes standardized criteria for hemolysis assessment [6][7].

2. Preparation/Acquisition

At baseline and every 30 minutes, three samples of blood will be collected from each loop and a blood smear will be prepared. The blood smear will be stained with a standard hematologic stain, such as Wright-Giemsa, and will be examined with brightfield microscopy.

Blood smear slides will be imaged using standardized acquisition settings to ensure consistency across timepoints. Imaging will be performed at 40x and 100x (oil immersion) magnification to evaluate cellular morphology and aggregation. All images will be acquired under constant illumination intensity, exposure time, and aperture. For each smear, five non-overlapping fields of view will be captured in a random sampling pattern. Images will be recorded at high resolution (2048 x 2048 pixels) and stored in an uncompressed format for downstream morphological analysis.

3. Analysis/Success Criteria

Analysis of the blood smear images will assess the integrity of blood cells and identify shear-related damage and immune activation. Visible hemolysis of RBCs, platelet density and clumping, and WBC morphology and clustering will be assessed. Success criteria will include hemolysis within acceptable limits with plasmal free hemoglobin < 20 mg/dl and NIH < 0.01 g / 100 L in accordance with ASTM F756 [7]. Platelet adhesion must remain < 5% surface coverage as measured by SEM analysis, and thrombus formation must remain < 10% device surface occlusion. RBC morphology must show <5% abnormal cells per field, and WBC activation must not show significant clustering compared to control conditions ($p > 0.05$).

SEM

1. Justification

Scanning electron microscopy (SEM) is used to characterize blood-device interactions on LVAD blood-contacting surfaces following flow-loop exposure. SEM enables high resolution visualization of platelet adhesion, platelet activation, leukocyte attachment, and early fibrous deposition, which are key indicators of device-induced thrombogenicity. This method supports surface level evaluation of thrombus formation as part of hemocompatibility testing for cardiovascular blood-contacting devices under ISO 10993-4 [4]. SEM provides nanometer-scale surface resolution, enabling direct visualization of cellular morphology and thrombus formation that cannot be resolved with light microscopy.

2. Preparation / Acquisition

After device exposure to the flow-loop for 6 hours, the LVAD will be disassembled and blood-contacting surfaces will be collected for SEM analysis. The device samples will be fixed in 2.5% glutaraldehyde for 24 hours at 4°C, followed by PBS rinsing. Samples will then undergo graded ethanol dehydration (30%, 50%, 70%, 90%, 100%), critical point drying, and sputter coating with a 5-10 nm gold layer prior to SEM imaging. SEM imaging will then be performed at 5-10 kV accelerating voltage, with magnification ranging from 500× to 5000×. The working distance will be maintained at 8-10 mm for an optimized resolution.

3. Analysis and Success Criteria

SEM will be used to assess the platelet adhesion, thrombus formation, cell morphology, aggregation, fibrous deposition, and leukocyte adhesion. The percent surface coverage and platelet density will serve as indicators of thrombosis. Success criteria will include <5% surface coverage, <10% surface occlusion, predominantly non-activated platelet morphology (absence of pseudopodia), no significant increase in adherent leukocytes compared to control device, and minimal fibrin deposition.

Micro-CT

1. Justification

Micro-CT has a high resolution of up to 50 micro-meters which is crucial when studying something as sensitive as clotting. Unlike X-ray and CT, micro-CT gives a 3D image of the LVAD which is helpful in understanding exact clotting locations and other areas prone to clotting. Further, 3D reconstruction enables formation of clotting mapping. Due to the difference in contrast created by the radioplaque contrast agents, quantifiable metrics like area and volume of clots can be assessed [8]. Another advantage of micro-CT is that it does not destroy or damage the sample.

2. Preparation

To simulate the left ventricle pressure-volume environment, a mock circulatory loop will be created according to ASTM F1841 [6] using either human blood at 37 degrees celcius. Ensure flow rate matches resting physiological range of 5 - 6 L/min [15] and pressure is 120/80 mmHg. Typical LVAD devices undergo 50 - 150 Pa [16] of shear stress and flow velocity is usually greater than 0.002 m/s [17]. These conditions should be incorporated in the mock circulatory loop shown in Figure 2. Faster velocity is better because stagnant blood may become a site for clotting which would lead to undesirable results. Similarly, shear stress outside the mentioned range may damage RBCs and lead to thrombosis. Once the loop has been set up, introduce appropriate radioplaque contrast agents like iodine, gold or barium sulfate-based agents and run LVAD at various speeds for about 4 hours. After testing is completed, treat the blood with PBS to remove residual blood from the LVAD.

Images will be acquired with a high-resolution X-ray system with voxel sizes 5 um-20 um. Scans will be performed at approximately 40-70 kVp and 100-200 uA with a full 360 degree rotation. Reconstruction will be completed using filtered back projection with artifact corrections.

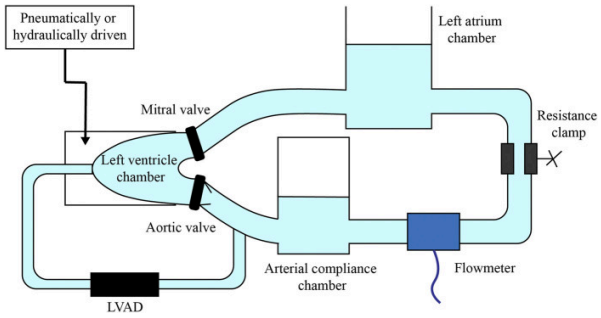


Figure 2 - Mock Circulatory Loop Setup [18]

3. Analysis and Success Criteria

To analyze the results, place each component of the LVAD under micro-CT at an appropriate resolution to get 3D grayscale images of the components. Using image segmentation, stack the grayscale images and create a 3D model of the LVAD. This 3D model will be instrumental in identifying qualitative information like presence of clot and areas prone to clotting and quantitative metrics such as surface coverage and hemolysis index. Success criteria for the aforementioned metrics have been summarized below in Table 2.

Table 2 - Acceptance Criteria for Quantitative Metrics for Micro-CT

Metrics	Acceptance Criteria	Standard Followed
Surface Coverage	< 10%	ISO 10933-4 [4]
Hemolysis Index	0 - 2%	ASTM-F756 [7]

In vivo animal study

The device will be explanted via thoracotomy in ten adult Dorset Sheep (40-60 kg) for 90 days [9]. All in vivo experiments will be conducted under Institutional Animal Care and Use Committee (IACUC) approval, in accordance with the US Animal Welfare Act and Public Health Service Policy on Humane Care and Use of Laboratory Animals. Animal care, study facilities, and protocols will adhere to 21 CFR 58 to ensure data integrity [19]. Furthermore, ISO 14708-5 will guide procedures specifically related to animal testing with ventricular assist devices [14]. Study procedures will include perioperative analgesia, continuous veterinary monitoring, and standardized humane endpoints to minimize animal distress. Additionally, baseline pump flow, power consumption and creatinine levels will be recorded. Animals will be humanely euthanized at study completion or upon reaching a predefined clinical endpoint using an IACUC-approved method, followed by confirmation of death prior to tissue harvesting for histological analysis [19].

Brightfield Microscopy

1. Justification

As outlined by FDA Guidance on *General Considerations for Animal Studies Intended to Evaluate Medical Devices*, if a device has a high likelihood of forming emboli, histology should be performed on all potential impacted tissues, with high-resolution and labeled images of all slides in a final study report. Assessments of inflammation, vascularization, and fibrin/thrombosis should be performed [20]. Brightfield microscopy enables standardized histopathological evaluation of these endpoints in LVAD-related hemocompatibility studies.

2. Preparation / Acquisition

Upon device explantation after 90 days, histological sections will be prepared at a thickness of 4 to 6 µm, and then imaged using a 4x objective for overview, 10x to 20x for quantitative assessment, and 40x for detailed morphology. Köhler illumination alignment will also be performed before imaging, and stage micrometer calibration will be conducted before each session. All images will be captured in 16 bit TIFF format to preserve data integrity. All outputs will include capsule thickness in µm, thrombus area percentage, and red blood cell morphology changes.

3. Analysis/Success Criteria

Brightfield microscopy will be used to evaluate thrombus formation, fibrin deposition, inflammatory response, and distal embolism in heart, kidney, liver, and lung tissue. A blinded veterinary pathologist will assess histology and will use a standardized scoring system for histological evaluation. Key success criteria include limited thrombus burden (<10% luminal occlusion), minimal fibrin deposition, no increase in inflammation, absence of embolism, and preservation of vascular integrity.

SEM

1. Justification

SEM is used post-explantation to evaluate chronic blood-device interactions on the LVAD surface after long-term implantation in the ovine model. This enables characterization of thrombus formation, fibrin layers, and platelet aggregation that occurs in physiologic, *in vivo* conditions. SEM will capture the cumulative effects of blood flow, biological response, and device wear over time, providing clinically relevant assessment of long-term thrombogenicity in accordance with ISO 10993-4 [4]. SEM's nanometer scale spatial resolution will enable detailed visualization of thrombi on the device surface as well as device degradation, such as microcracks and pitting, over long-term implantation.

2. Preparation/ Acquisition

At the study endpoint, the LVAD will be explanted from the ovine model. Blood-contacting surfaces of the LVAD will be collected, fixed, and sputter-coated for SEM imaging.

For scanning electron microscopy (SEM), the accelerating voltage will range between 5-15 kV with magnification between 100x and 5000x. When it is required, the sputter coating thickness will also be recorded and elemental analysis will be performed if particulates are observed, using EDS. The surface defect density per mm^2, platelet activation morphology, and evidence of coating delamination or micro pitting will also be captured for each image.

Gross photographic documentation will be completed prior to any processing in order to preserve visual evidence of the surface conditions. The inflow cannula and outflow graft regions will then be sectioned but will still preserve the tissue device interface. The samples will be fixed in formalin for histology, and a separate surface sampling for SEM will be conducted without disrupting thrombus deposits. Throughout this process, chain of custody documentation will be maintained to ensure traceability and regulatory compliance.

3. Analysis and Success Criteria

Evaluation will include platelet adhesion and activation morphology, fibrin deposition, leukocyte attachment, and thrombus structure, especially at the inflow cannula and outflow graft. Quantitative analysis will be performed with standardized image analysis workflows to determine percent surface coverage, thrombus burden, and localized surface degradation. Success criteria will include <5% surface coverage, <10% surface occlusion, predominantly non-activated platelet morphology (absence of pseudopodia), no significant increase in adherent leukocytes compared to control device, minimal fibrin deposition, and surface integrity (lack of microcracks/pitting).

CT

1. Justification

Cardiac computed tomography (CT) angiography is selected for in-vivo evaluation of LVAD function and safety due to its ability to provide high-resolution, three-dimensional visualization of device geometry and vasculature. This enables assessment of device positioning, including inflow cannula and outflow graft alignment in the ovine model [12][13]. CT can effectively detect thrombus formation because it can detect contrast filling defects, or areas where a clot would displace the radiopaque contrast agent. Furthermore, it can also identify mechanical characteristics such as graft kinking, malpositioning, and obstruction. Volumetric imaging allows for quantitative analysis of device occlusion, cross-sectional area, and thrombus volume in the complex geometry. Additionally, CT is also used pre-implantation to map animal anatomy and ensure compatibility with the device. ECG-gated CT acquisition can be utilized to minimize motion artifacts in the heart to clarity of imaging.

The use of CT as a clinical imaging modality aligns with ISO 14708, which specifies safety and performance requirements for active implantable medical devices, with part 5 addressing circulatory support devices. This standard outlined requirements for evidence of stable anatomic positioning, absence of thrombus, and effectiveness of flow, which can be directly assessed with cardiac CT angiography [14].

2. Preparation/ Acquisition

Cardiac CT angiography will be performed with a 192 slice multidetector CT scanner for large animal imaging. A pre-implantation scan will be conducted for assessment of ovine subject cardiac anatomy prior to toracotomy. Imaging will be conducted at baseline, 1 week (acute), 1 month (subacute), and 3 months (chronic) [21]. Imaging will be conducted under anaesthesia to minimize motion artifacts, with ECG-gated acquisition will be used to account for cardiac motion. Scans will be acquired at a tube voltage of 100-120 kVp and a tube current of 200-400 mA, with automatic exposure control based on animal size. A slice thickness of 0.5-0.75 mm will be used to achieve high spatial resolution needed to evaluate the LVAD inflow cannula, outflow graft, and thrombus formation. To optimize temporal resolution for cardiac imaging, a low pitch and fast gantry rotation time will be utilized. Intravenous contrast will be administered through a central venous catheter at a rate of 3-5 mL/s with a total volume of 1-2 mL/kg, followed by saline [22]. Three-dimensional reconstructions will be generated for quantitative analysis of thrombus formation, device positioning, and graft geometry.

3. Analysis and Success Criteria

The qualitative indicator for clot will be the presence of thrombi on any LVAD components which will be seen using contrast agents using CT. Regions without the contrast agents will be investigated as regions prone to clotting. Success will be defined as a lack of detectable filling defects, especially the inflow cannula, pump graft, and outflow graft. Additionally, an absence of graft narrowing or kinking should be observed by the graft maintaining >95% of its diameter throughout the study. To prevent suction events, the alignment angle of the inflow cannula must remain within 15 degrees of the ventricular long axis and at least 1.5 cm from the ventricular septum [23].

For functional success, three quantitative metrics can be used. If pump flow is maintained within normal range of 2-5 L/min, it shows there was no change in velocity due to an obstruction. Similarly, for power consumption any unusual trend indicates towards abnormal obstruction. The reason creatinine levels are measured is to analyze if there are effects of a potential clot on kidneys.

Acquisition Parameters:

With the processing protocols established, the broader preclinical and clinical testing model can then be considered. Preclinical evaluation will begin with in vitro hemocompatibility testing under ISO 10993-4 and ASTM F756 [4], [6], [7]. If results remain within acceptable limits, large animal implantation will be conducted in order to assess device performance under physiologic flow conditions [9]. Microscopy endpoints will focus on thrombus formation,

platelet adhesion, surface degradation, and tissue inflammatory response, with the final hemocompatibility endpoints.

Finally, cost considerations must also be acknowledged within the overall testing framework. LVAD therapy represents a significant medical investment, with an estimate that first year LVAD therapy costs range between approximately $190,000 and $220,000, including hospitalization and device related care [24]. Looking at the device only, costs typically fall between $80,000 and $100,000, depending on institution and payer. In comparison, microscopy related costs primarily include histology reagents, immunostaining supplies, SEM and confocal instrument usage, animal study surgical & housing expenses, and personnel time for imaging and analysis. Among these, animal implantation studies represent the largest cost component within the microscopy validation timeline.

Data Management Plan

Concerns	Solutions
Data Collection/Creation, Storage and Back Up	Since brightfield microscopy requires blood to be sampled, histology slides would have to be created. An image of each stained section will be captured using the EVOS microscope and saved. Similarly, for SEM, the observed cells will be captured using an Field Emission SEM microscope and saved. Images captured via the EVOS and Field Emission SEM microscopes will be transferred via a secure, encrypted local area network (LAN) directly to Abbott's centralized enterprise servers. For CT imaging, all volumetric datasets (in a DICOM format) will be directly exported from the CT scanner to Abbott's secure enterprise servers via encrypted PACS integration. Raw projection data, reconstructed volumes, and processed segmentation files will all be retained. All CT datasets will include associated reconstruction parameters and contrast agent details to ensure reproducibility. All data will be stored on secure, FDA-compliant enterprise servers maintained by Abbott. Data will be managed using validated electronic data capture (EDC) systems compliant with 21 CFR Part 11 [25]. Access will be role-based and controlled through secure login authentication.

	All labelling of images will be done as follows: • **Technique_Sample.No_UniqueID (if any)** • *Brightfield Example:* Brightfield_1_XYZ789 • *SEM Example:* SEM_Surface1_Week1_XYZ789 • *CT Example:* CT_30Day_LVAD01_XYZ789 All files will follow standardized naming conventions to ensure traceability across experiments, timepoints, and imaging modalities, supporting reproducibility and regulatory review.
Access and Security	Access to imaging data will be controlled through Abbott's secure, and FDA compliant enterprise data management systems. All data will be stored on validated internal servers that comply with 21 CFR Part 11 requirements for electronic records [25]. Access will be role-based, with permissions granted based on the person's responsibilities. Along with this, having multi factor authentication and audit trails implementation will ensure data integrity, traceability, and regulatory compliance. All data access, modification, and analysis activities will be logged and monitored to maintain full transparency and accountability throughout the study lifecycle.
Metadata	All techniques must be accompanied with metadata. Some crucial information to include is: • Date of Experiment • Operator/Analyst ID • Protocol Number and ID followed • Experiment information (numerical aperture, magnification, pixel size, temperature, humidity, what microscope/machine used) • Cell used • Stain/Dye used • Implant name and type • Implant location For CT imaging, additional metadata will include:

	• Scanner model and manufacturer • Tube voltage (kVp) and current (mA) • Slice thickness and voxel size • Reconstruction algorithm/kernel • Contrast agent type and concentration • Injection rate and timing (for in vivo studies) • Cardiac/respiratory gating parameters (if applicable)
Data Breach/IP/Copyright Infringement Issues	All data will be managed in accordance with Abbott's internal data governance policies and FDA 21 CFR Part 11 requirements [25]. *Data Breach*: If a data breach happens, access to all the affected systems will be immediately restricted. This incident will then be reported to Abbott's Information Security and IT teams, and a formal investigation will be conducted. Any compromised credentials will be revoked and reissued, and all system activity logs will be audited to identify the source and scope of the breach. Data will not be stored on any external or removable devices to minimize risk. *IP Breach* - Before onboarding, all contributors in any capacity (student, doctoral candidate, full-time scientist) will be required to sign a Non-Disclosure Agreement drafted by the patent office on-site. For the same reason, no contributor should be permitted to click pictures on a personal cell-phone. To click pictures, lab cameras should be saved to the lab desktop, uploading it to the server. In the event of any IP breach, the legal office will be consulted for the next steps. Confidential information will be restricted to only the doctoral, full-time scientists and PI. *Copyright Infringement* - All images used will be originally created and captured. No external, online images will be used. No AI-generated images will be used too. The patent office on-campus will be consulted regarding the licensing of the image. All experimental procedures, observations, and imaging data will be recorded in validated

	electronic laboratory notebook (ELN) systems integrated in Abbott's data infrastructure. Each user will maintain individual, time stamped entries linked to their unique ID. The ELN system will ensure secure and real time data entry, automatic version control, audit trails for all modifications, and centralized access for authorized personnel. This approach will ensure full traceability, transparency, and compliance with regulatory documentation standards.
Long-term Data Plan	All data will be stored in accordance with Good Laboratory Practice (GLP) standards and FDA regulatory requirements. Imaging and associated metadata will be securely stored for a minimum of 10 years after PMA. Data will be also maintained on validated enterprise servers with multiple backup systems. At the end of the retention period, data will be securely destroyed in compliance with the institutional and regulatory policies, such as permanent deletion from active and backup storage systems, verification audits to ensure complete removal, and/or secure decommissioning of any associated hardware All data lifecycle actions will be documented to ensure compliance and traceability.

Data Analysis & Interpretation:

To ensure scientific validity, reproducibility, and regulatory defensibility, all imaging data will be analyzed using standardized and validated procedures by qualified personnel with clearly defined roles. Templates for regions of interest will be applied uniformly across all samples, and acquisition settings will remain locked within each imaging batch to prevent any variability. Qualitative inflammatory grading will then be performed using blind scoring to reduce bias, and at least ten percent of samples will undergo secondary reviewer verification. Using ANOVA, statistical analysis will be conducted with post hoc testing and a significance threshold level of $p < 0.05$. Finally, hemolysis results will be evaluated according to ASTM F756 thresholds, and all raw and processed image files will be archived and stored per the Data Management Plan [7].

Histological analyses will be performed by board certified pathologists with experience in cardiovascular tissue evaluation. Brightfield microscopy data (such as RBC morphology,

WBC activation, and platelet aggregation) will be reviewed and scored using standardized histological grading criteria [26]. SEM analyses will be conducted by trained biomedical engineers with expertise in electron microscopy and biomaterial surface characterization. SEM images will be evaluated for platelet adhesion, thrombus formation, fibrin network development, and surface degradation. Statistical analyses will be performed by trained researchers with experience in biomedical data analysis using software such as MATLAB, Python, or GraphPad Prism [27].

Hemolysis will be assessed by calculating the percentage of damaged or fragmented red blood cells per field of view and comparing to controls [28]. SEM images will be used to quantify platelet surface coverage (%) and thrombus density per mm^2 using image segmentation techniques. Then, inflammatory response will be evaluated based on white blood cell morphology and clustering across defined timepoints. Finally, regions of interest (ROI's) will be applied uniformly across all samples to ensure comparability, and all results will be analyzed using a one-way ANOVA with post hoc testing, with the statistical significance defined as $p < 0.05$.

To minimize any biases and ensure reliability, a blinded analysis will be implemented where applicable, and at least 10% of samples will undergo an independent secondary review [29]. Imaging parameters and analysis pipelines will remain fixed within each dataset to prevent variability. All results will be directly compared to predefined success criteria, including ASTM F756 hemolysis thresholds and platelet adhesion limits, to determine whether the device meets safety and performance requirements before progressing to the next testing phases.

CT datasets will be analyzed using validated image processing software such as Mimics, 3D Slicer. 3D reconstructions will then be used to quantify thrombus volume (mm^3), percent occlusion (%), and flow path obstruction within the LVAD and associated graft structures. The segmentation protocols will be standardized across all datasets, and inter observer validation will be performed on at least 10% of samples to ensure reproducibility.

Timeline:

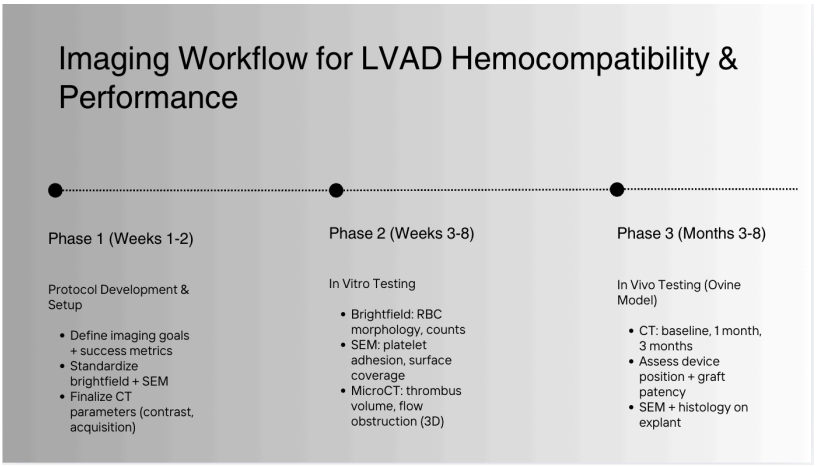


Figure 3. Timeline of imaging-based evaluation of LVAD performance integrating brightfield microscopy, SEM, and CT across in vitro, preclinical, and clinical phases.

The microscopy component of this testing plan will be carried out in sequential phases so that hemocompatibility, thrombogenicity, and tissue response can be evaluated in a logical and controlled manner. Rather than performing all analyses simultaneously, the approach will progress from controlled in vitro studies to preclinical validation and finally to clinical and post market evaluation.

Phase 1, the protocol finalization stage, will be completed within weeks 1-2. During this time, the specific microscopy modalities to be used will be finalized, and will include brightfield light microscopy for cellular level blood analysis and SEM for high resolution surface characterization of device-material interactions. At the same time, the exact hemocompatibility endpoints, required under ISO 10993-4 [4], [7] will be finalized to hemolysis (plasma free hemoglobin and percent hemolysis), platelet adhesion (surface coverage %), thrombus formation (percent occlusion), and inflammatory response (white blood cell morphology and clustering). In addition, standard operating procedures for blood handling, fixation, and imaging preparation will be established. Calibration procedures and acquisition reproducibility will also be validated to ensure consistency and reliability before formal testing begins. In addition, CT imaging protocols will be finalized, including contrast agent selection, acquisition parameters, and reconstruction workflows for both in vitro and in vivo imaging.

Phase 2 will involve in vitro hemocompatibility testing and will take place during weeks 3-8. Once the protocols have been finalized, hemolysis testing will be performed in accordance with ASTM F756 [7]. Additional blood material interaction studies will also be conducted to further evaluate thrombogenic behavior. Hemocompatibility metrics defined under ISO 10993-4

[4], [7] will then be quantified and compared against appropriate positive and negative control materials. The hemocompatibility metrics will include percent hemolysis, platelet surface coverage (%), thrombus density per mm^2 , and white blood cell activation and clustering. Micro CT imaging will be performed on LVAD components after flow loop exposure to quantify early thrombus formation and flow obstruction in three dimensions. This stage will serve as the first formal safety checkpoint before progressing to animal implantation.

Phase 3 will consist of preclinical in vivo validation and is expected to occur between months 3-8. If the in vitro results meet acceptable thresholds, the device will be implanted in a large animal model. The in vivo model will use adult Dorset sheep (40-60 kg), as this model is standard for LVAD evaluation because of the ovine having a comparable cardiovascular physiology to humans [9]. Then, the device explantation and imaging will occur at 1 week (acute response), 1 month (subacute response), and 3 months (chronic response) to allow for detailed imaging of device surfaces and tissue device interfaces. CT imaging will then be done at baseline, 1 month, and 3 months post implantation non invasively. At these stages, thrombus formation, inflammatory response, and surface integrity will all be assessed under physiologic flow conditions to evaluate the biological performance before clinical use.

Throughout development, microscopy results will guide safety and design decisions at defined checkpoints. During the early in vitro stage, hemolysis values will be evaluated against ASTM F756 thresholds [7] to determine whether red blood cell damage remains within acceptable limits. The key evaluation metrics of these images are as follow: hemolysis is $< 2\%$, platelet adhesion is $< 5\%$ surface coverage, thrombus formation is $< 10\%$ occlusion, inflammatory response (no significant increase vs control ($p > 0.05$)), and device integrity (no structural degradation observed). As testing progresses, platelet adhesion and fibrin deposition will be assessed, relative to the control materials, to evaluate thrombogenic potential. During the preclinical stage, explanted devices will be examined for thrombus formation and inflammatory response at the tissue device interface.

Together, these checkpoints allow early identification of hemocompatibility concerns and support corrective action before regulatory submission or broader clinical implementation.

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